Binomial Distribution

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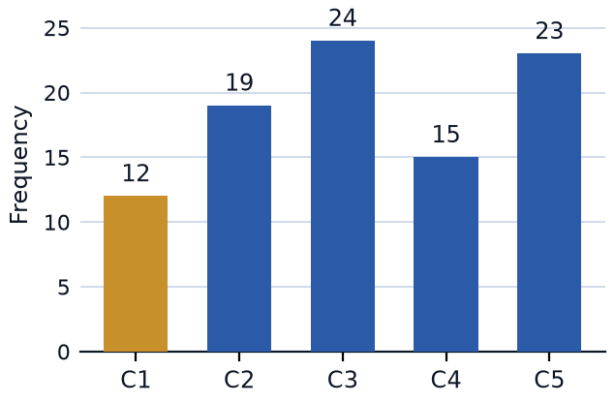
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Conditions for a binomial distribution

State the four conditions required for a random variable to follow a binomial distribution.

2 [Foundational] • Target: Conditions for a binomial distribution

A survey asks 50 people whether they support a new policy, and X is the number who say yes, assuming each person's response is independent with the same probability of support. Explain why X follows a binomial distribution.



3 [Foundational] • Target: Conditions for a binomial distribution

A researcher rolls a die 15 times and counts the number of times a 3 appears. Explain why this count follows a binomial distribution.

4 [Foundational] • Target: Calculating binomial probabilities

A fair coin is tossed 8 times. Find the probability of getting exactly 5 heads, using $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

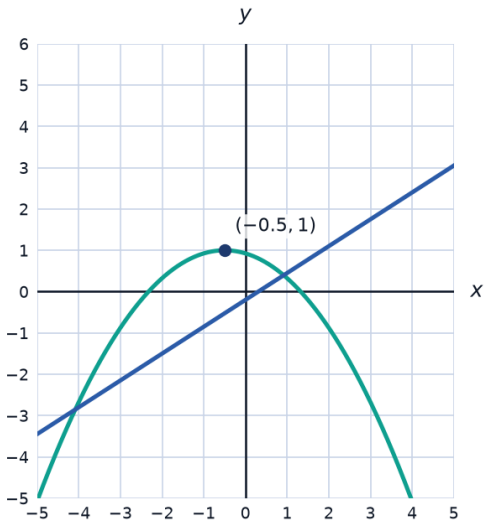
5 [Foundational] • Target: Calculating binomial probabilities

A factory produces items with a defect rate of 10%. In a sample of 6 items, find the probability that exactly 2 are defective.

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6 [Foundational] • Target: Calculating binomial probabilities

A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.



7 [Foundational] • Target: Calculating binomial probabilities

A factory produces items with a defect rate of 20%. In a sample of 5 items, find the probability that exactly 1 is defective.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Cumulative binomial probabilities

A biased coin has $P(\text{heads}) = 0.3$. It is tossed 5 times. Find $P(X \leq 1)$, where X is the number of heads.

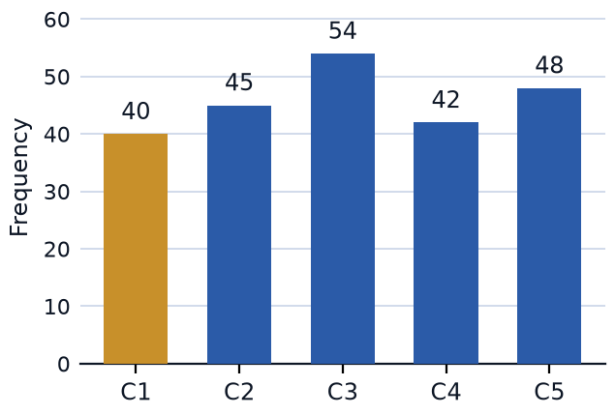
9 [Intermediate] • Target: Cumulative binomial probabilities

Using the same distribution $X \sim B(5, 0.3)$, find $P(X \geq 2)$.

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10 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(40, 0.25)$. Find the mean and variance, using $E(X) = np$ and $\text{Var}(X) = np(1-p)$.



11 [Intermediate] • Target: Mean and variance of a binomial distribution

A random variable $X \sim B(60, 0.2)$. Find the mean and variance.

12 [Intermediate] • Target: Finding an unknown parameter

A random variable $X \sim B(n, 0.4)$ has mean $E(X) = 20$. Find n.

13 [Intermediate] • Target: Finding an unknown parameter

A random variable $X \sim B(n, 0.25)$ has variance $\text{Var}(X) = 9$. Find n.

14 [Intermediate] • Target: Using the binomial distribution to model a real scenario

A multiple-choice quiz has 10 questions, each with 4 options, only one of which is correct. A student guesses randomly on every question. Find the probability the student gets at least 8 questions correct.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

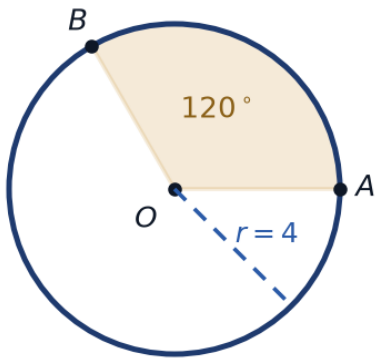
15 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 14.01 Energy, Work & Power

A basketball player has a 70% free-throw success rate. Find the probability that the player makes exactly 3 out of 5 free throws.

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16 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 15.01 Representation of Data

An archer hits the target with probability 0.8 on each shot, independently. Find the probability that the archer hits the target exactly 4 times out of 5 shots.



17 [Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 16.01 Permutations & Combinations

Explain why drawing 5 cards from a standard deck WITHOUT replacement and counting the number of aces drawn is NOT a binomial distribution, while drawing 5 cards WITH replacement would be.

18 [Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 17.01 Discrete Random Variables

Explain why the number of sixes rolled in 10 rolls of a fair die IS a binomial distribution, while the number of rolls needed to get the first six is NOT.

19 [Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 19.01 Normal Distribution

Explain why the number of heads obtained by flipping a fair coin exactly 20 times IS a binomial distribution, while the number of flips needed to get 3 heads is NOT.

20 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

This question has two parts. A machine produces bolts, of which 5% are defective. A quality inspector randomly samples 20 bolts each hour. (a) Find the expected number of defective bolts in a sample of 20. (b) The inspector rejects the whole batch if 3 or more defective bolts are found in the sample. Find the probability that the batch is rejected, given $P(X = 0) \approx 0.3585$, $P(X = 1) \approx 0.3774$, and $P(X = 2) \approx 0.1887$ for $X \sim B(20, 0.05)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Conditions for a binomial distribution

A machine produces metal rods whose lengths are normally distributed with unknown mean μ cm and known standard deviation $\sigma = 0.8$ cm. A quality control manager tests the hypotheses $H_0 : \mu = 25.0 \quad \text{against} \quad H_1 : \mu < 25.0$ at the 5% level of significance. A random sample of 16 rods yields a sample mean length of $\bar{x} = 24.6$ cm. Given that the critical value for this test is $z_{\text{crit}} = -1.645$, which of the following gives the correct test statistic z and the corresponding conclusion?

a. $z = -2.0$; reject H_0 because $-2.0 < -1.645$

b. $z = -2.0$; do not reject H_0 because $-2.0 < -1.645$

c. $z = -0.5$; do not reject H_0 because $-0.5 > -1.645$

d. $z = -8.0$; reject H_0 because $-8.0 < -1.645$

22 [Foundational] • Target: Calculating binomial probabilities

The unique real root α of the equation $x^3 + 2x - k = 0,$ where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$

b. $0 < k < 2\sqrt{2}$

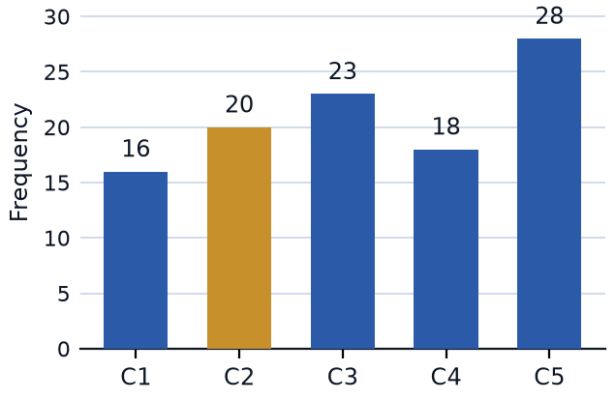
c. $0 < k < 3$

d. $0 < k < 12$

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23 [Foundational] • Target: Cumulative binomial probabilities

A random sample of $n = 50$ observations is taken from a population. The sample values are summarised as follows: $\sum x = 450, \quad \sum x^2 = 4295$. Find an unbiased estimate of the population variance.



a. 5.0

b. 4.9

c. 2.24

d. 9.0

24 [Foundational] • Target: Mean and variance of a binomial distribution

A machine produces steel rods whose lengths are normally distributed with standard deviation $\sigma = 2.4$ cm. A quality controller suspects that the mean length has increased from the target value of 50 cm. A random sample of 36 rods is measured, yielding a sample mean of $\bar{x} = 50.8$ cm. Find the value of the standardized test statistic z for testing $H_0 : \mu = 50$ against $H_1 : \mu > 50$.

a. 2.0

b. 0.33

c. 1.2

d. 0.5

25 [Foundational] • Target: Finding an unknown parameter

Formulate the null and alternative hypotheses for the following scenario: "A factory claims the mean weight of a product is 250g. An inspector wants to test if it has changed (either direction)."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 3,\ H_1 : \mu > 3$

c. $H_0 : \mu = 10,\ H_1 : \mu < 10$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

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26 [Foundational] • Target: Using the binomial distribution to model a real scenario

Formulate the null and alternative hypotheses for the following scenario: "A manufacturer claims the mean lifetime of a battery is 500 hours. A researcher suspects it is less than this."

a. $H_0 : \mu = 500,\ H_1 : \mu < 500$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 3,\ H_1 : \mu > 3$

d. $H_0 : \mu = 250,\ H_1 : \mu e 250$

27 [Foundational] • Target: Distinguishing binomial situations from non-binomial situations

Formulate the null and alternative hypotheses for the following scenario: "A school claims the mean test score is 70. A teacher believes it is actually higher."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

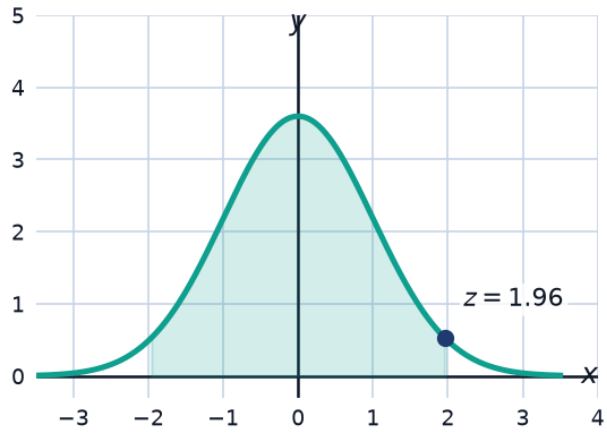
b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$

28 [Foundational] • Target: A well-designed word problem: quality control sampling

A confidence interval for the mean uses $z = 1$, $\sigma = 4$, $n = 9$. Find the margin of error $E = z\sigma/\sqrt{n}$.



a. $\frac{7}{3}$

b. $\frac{4}{3}$

c. $\frac{17}{12}$

d. $\frac{8}{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Conditions for a binomial distribution

A sample of size $n = 9$ is drawn from a population with mean $\mu = 42$ and $\sigma = 8$. Given $\bar{x} = \frac{118}{3}$, find $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$.

a. 0

b. 1

c. -1

d. -3

30 [Intermediate] • Target: Calculating binomial probabilities

A sample of size 6 has $\sum x = 282$ and $\sum x^2 = 13312$. Find the unbiased estimate of the population variance.

a. $\frac{6656}{3}$

b. $\frac{58}{5}$

c. $\frac{63}{5}$

d. $\frac{29}{3}$

31 [Intermediate] • Target: Cumulative binomial probabilities

$X \sim Po(3)$ and $Y \sim Po(3)$ are independent. Find the parameter of the distribution of $X + Y$.

a. 9

b. 0

c. 6

d. 7

32 [Intermediate] • Target: Mean and variance of a binomial distribution

A sample of size 6 gives values: 5, 12, 18, 11, 17, 18. Find the sample mean $\bar{x}$.

a. 18

b. $\frac{29}{2}$

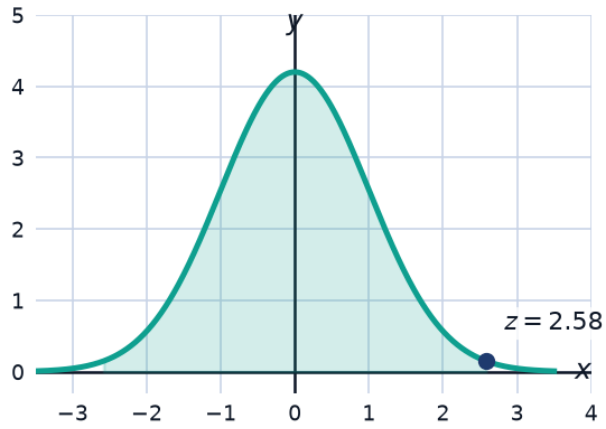
c. $\frac{27}{2}$

d. $\frac{81}{7}$

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33 **[Intermediate]** • **Target: Finding an unknown parameter**

A confidence interval for the mean uses $z = 1$, $\sigma = 8$, $n = 9$. Find the total width of the confidence interval, $2E$.



a. $\frac{19}{3}$

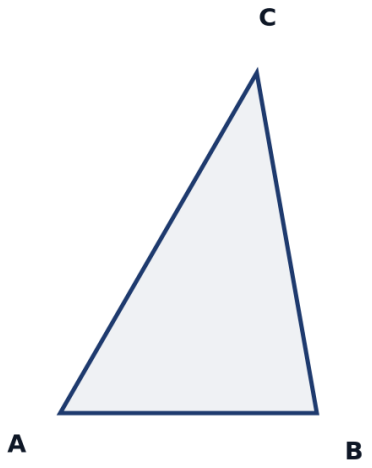
b. $\frac{16}{3}$

c. 8

d. $\frac{8}{3}$

34 **[Intermediate]** • **Target: Using the binomial distribution to model a real scenario**

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.



a. 5.7

b. 6.7

c. 7.7

d. 4.7

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35 **[Intermediate]** • **Target: Distinguishing binomial situations from non-binomial situations**

The polynomial $P(x)$ is defined by $P(x) = (2 + ax)^n + bx^4 + cx - 4$ where n is an integer greater than 2, and a, b, c are constants with $a > 0$. In the binomial expansion of $(2 + ax)^n$ in ascending powers of x: - the coefficient of x^2 is twice the coefficient of x, - the coefficient of x^3 is equal to the coefficient of x^2. Given that $(x + 2)$ is a factor of $P(x)$, and that the remainder when $P(x)$ is divided by $(x + 1)$ is 7, find the remainder when $P(x)$ is divided by $(x - 1)$.

a. 1011

b. 1031

c. 1015

d. 499

36 **[Intermediate]** • **Target: A well-designed word problem: quality control sampling**

Six-digit numbers are formed using digits chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with no digit repeated in any number. Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.

a. 8160

b. 7680

c. 9120

d. 2880

37 **[Intermediate]** • **Target: Conditions for a binomial distribution**

The real numbers x and y satisfy the simultaneous equations: $\log_3(x) + 2\log_9(y - 1) = 2$ $4^y \cdot 8^{-\frac{x}{3}} = 32$ Find the value of $x^2 + y^2$.

a. 25

b. 10

c. $\frac{145}{4}$

d. 50

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Calculating binomial probabilities • Cross-Lesson: 17.01 Discrete Random Variables

In the expansion of $(1+ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a>0$. A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x-a)^2$ as a factor and leaves a remainder of n when divided by $(x-1)$. Find the remainder when $P(x)$ is divided by $(x+a)$.

a. 48

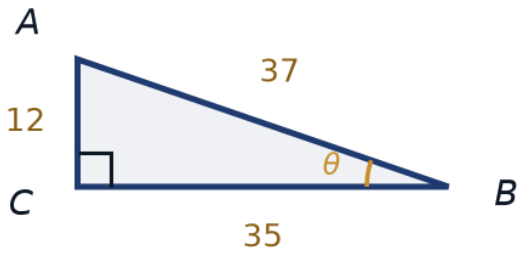
b. -48

c. -112

d. 0

39 [Challenging] • Target: Cumulative binomial probabilities • Cross-Lesson: 19.01 Normal Distribution

Let the complex number z be represented in polar form as $z=r(\cos\theta+i\sin\theta)$. Suppose z satisfies the equation $z^3=-64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2}<\arg(z)<2\pi$), find the real part of z.



a. 4

b. -4

c. $2\sqrt{3}$

d. $-2\sqrt{3}$

40 [Challenging] • Target: Mean and variance of a binomial distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Determine the complete set of real values of the constant k for which the equation $2\cos 2\theta-4\sin\theta=k$ has at least one real solution for θ.

a. $-6\le k\le 3$

b. $-6\le k\le 2$

c. $k\le 3$

d. $-3\le k\le 6$

41 [Challenging] • Target: Finding an unknown parameter • Cross-Lesson: 3.01 Graphs & Transformations

Find the set of all real solutions to the equation $\log_2(x-1)-\log_4(x+14)=1$

a. $x=11$ only

b. $x=11$ and $x=-5$

c. $x=5$ only

d. $x=16$ only

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42 [Challenging] • Target: Using the binomial distribution to model a real scenario • Cross-Lesson: 4.01 Coordinate Geometry

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

43 [Challenging] • Target: Distinguishing binomial situations from non-binomial situations • Cross-Lesson: 5.01 Trigonometry

The complex number z is defined by $z = \frac{a + 3i}{2 + ai}$, where a is a real constant. Given that $\arg(z) = \frac{\pi}{4}$, find the value of a.

a. 1

b. -6

c. 1 or -6

d. -1 or 6

44 [Challenging] • Target: A well-designed word problem: quality control sampling • Cross-Lesson: 6.01 Differentiation & Integration

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. 100

b. 110

c. 10

d. 210

45 [Challenging] • Target: Conditions for a binomial distribution • Cross-Lesson: 7.01 Statistics & Probability

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

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